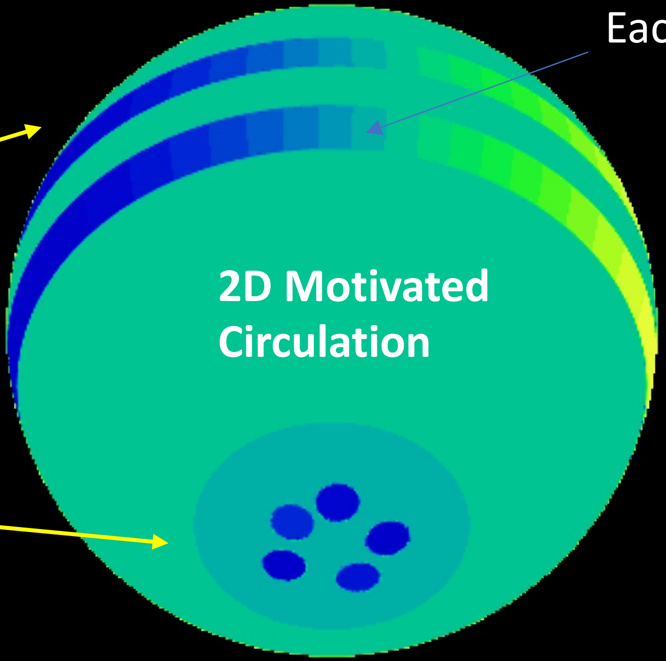
A Polar Vortex Forward-Modelling Framework: MAYMO 2D + 1D

Combining state-of-the-art models into a simpler multi-component framework

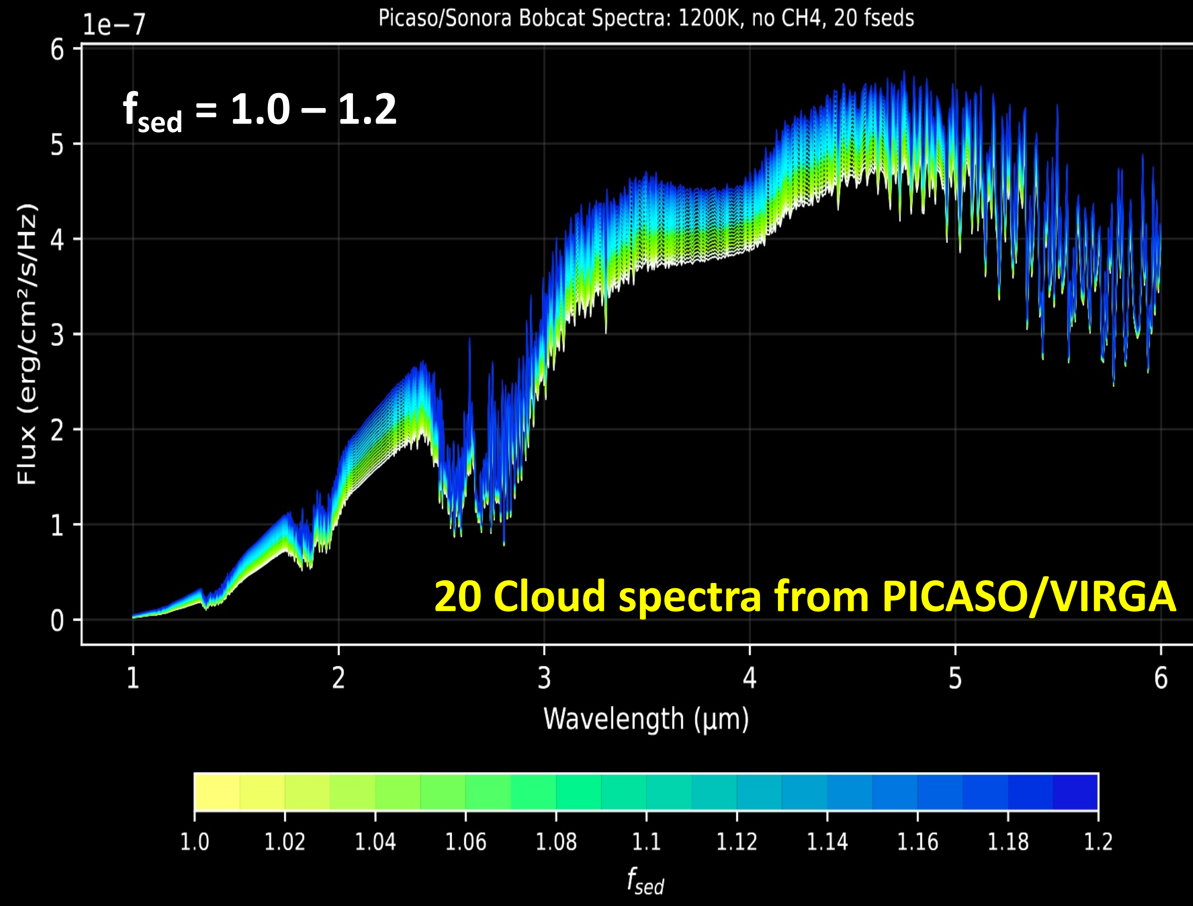
(Fuda in prep. 2026)

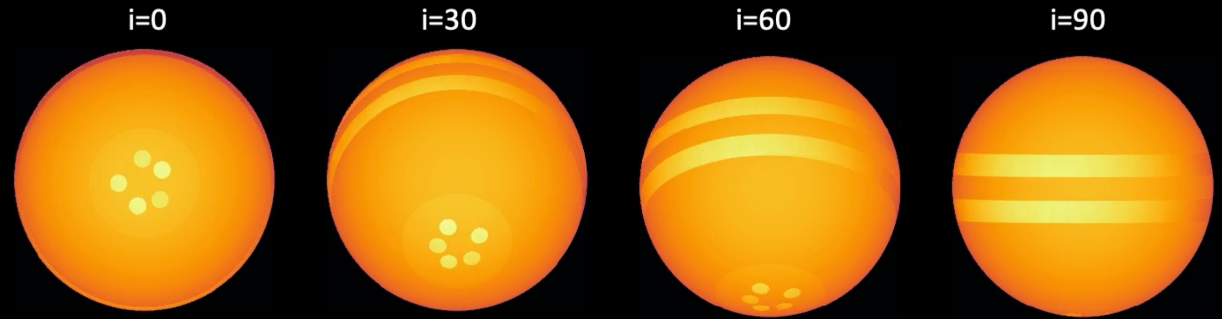
Each of these is a spectra!

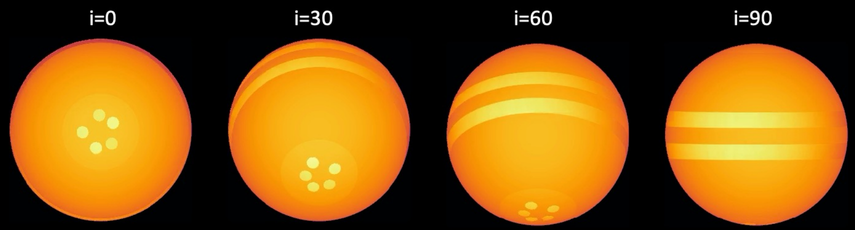
Short-period Bands (planetary-scale waves)

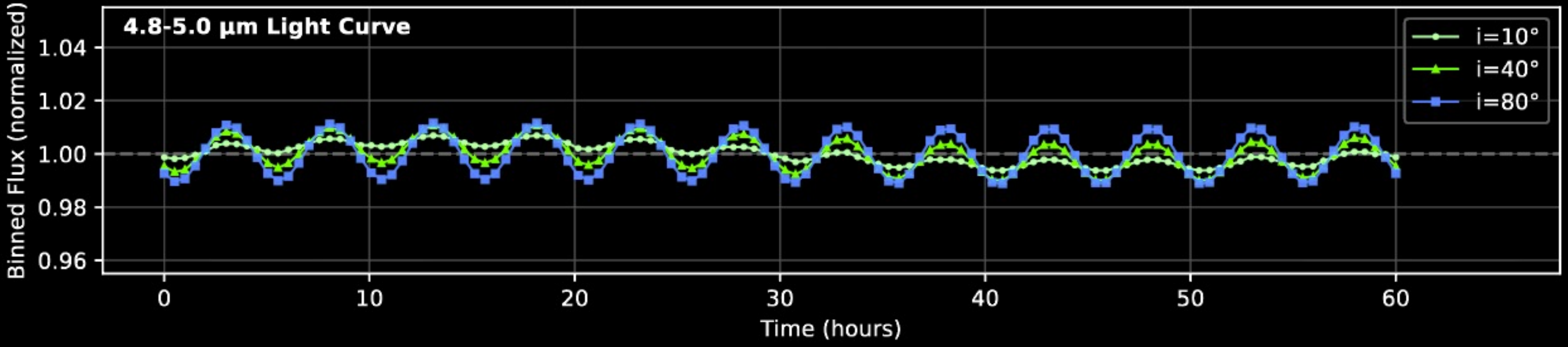


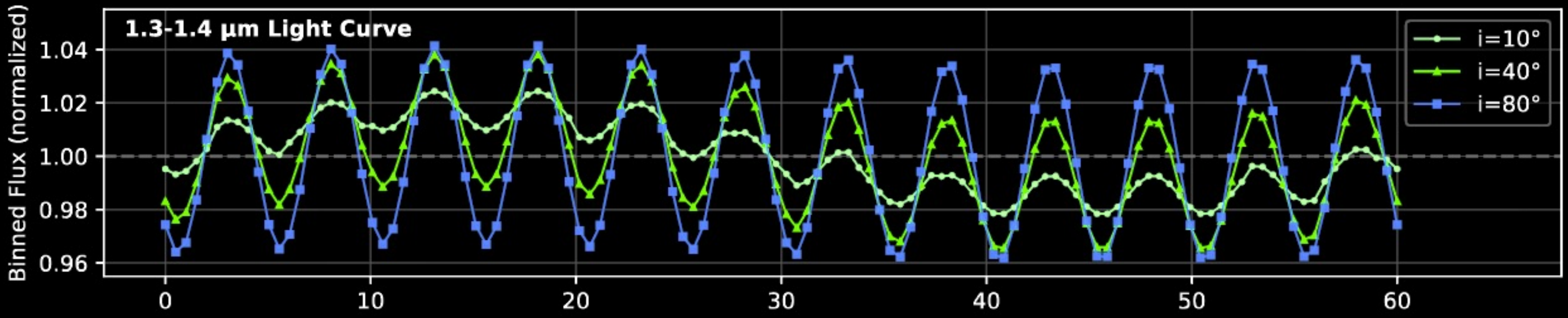
Long-Period Polar Vortices









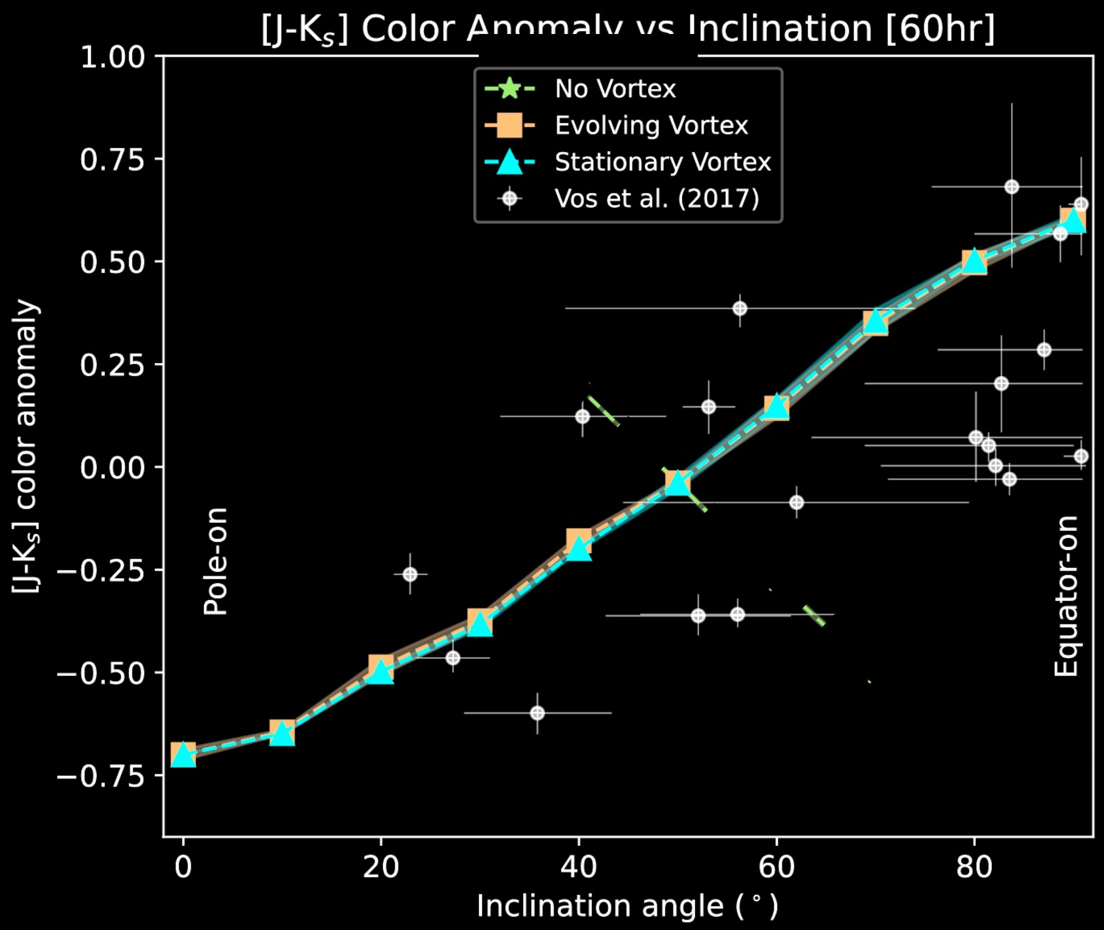


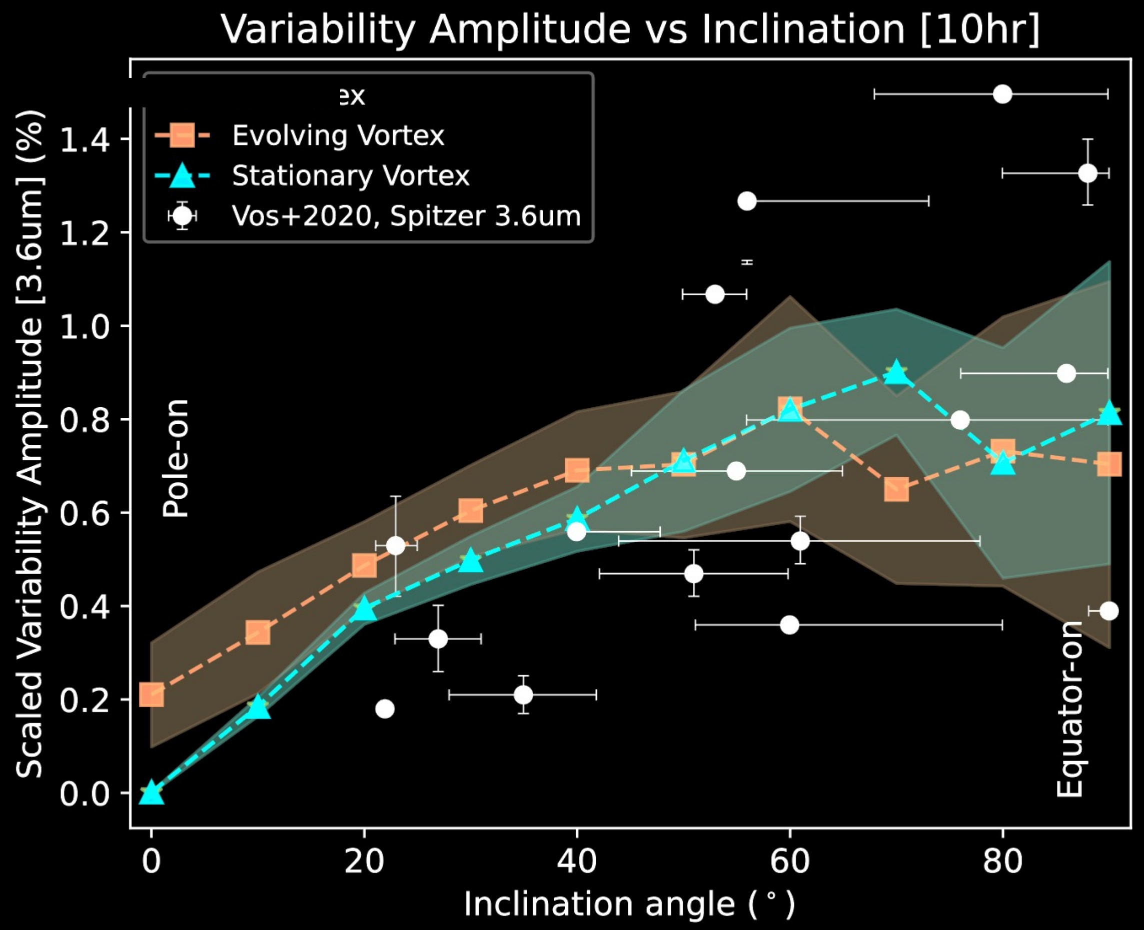
Nguyen Fuda - Pole-to-equator Divergences

Simulated spectral timeseries consistent with observations

Variability & colors vs inclination trends: consistent with field BDs (Vos et al. 2020)

(Will have updated version in Fuda et al. 2026 in prep)





Color v. inclination **can be explained with pole-to-equator cloud difference**

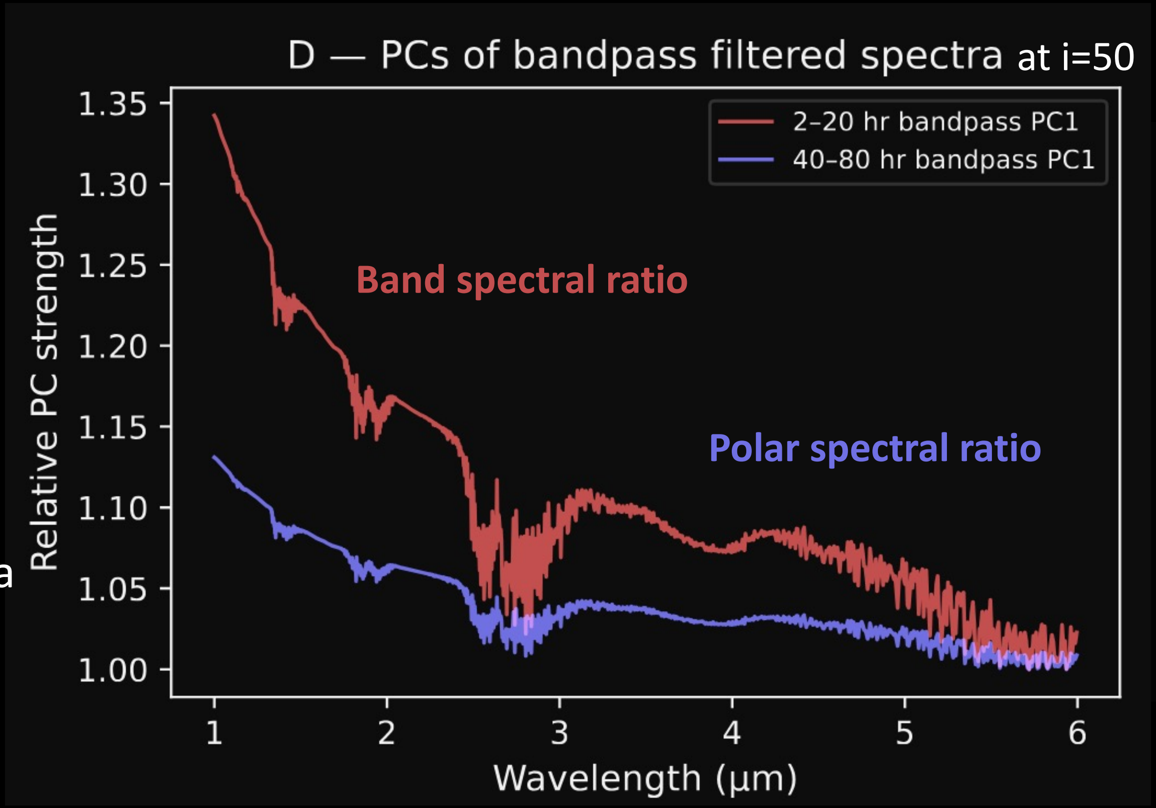
(Fuda & Apai 2024)

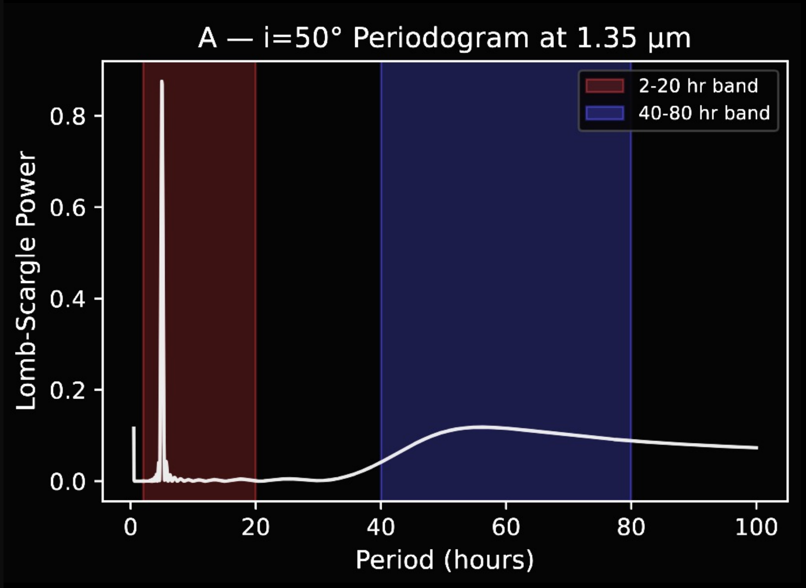
Variability v. inclination **points to higher amplitude from the equatorial atmosphere**

Simulated spectral timeseries w/ Maymo

(Fuda in prep. 2026)

i=60





Spectral ratio = spectra at max / min flux

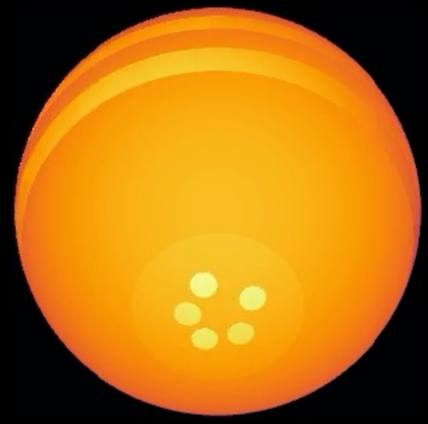
Polar ratio less than bands because less of it is visible at i=50.

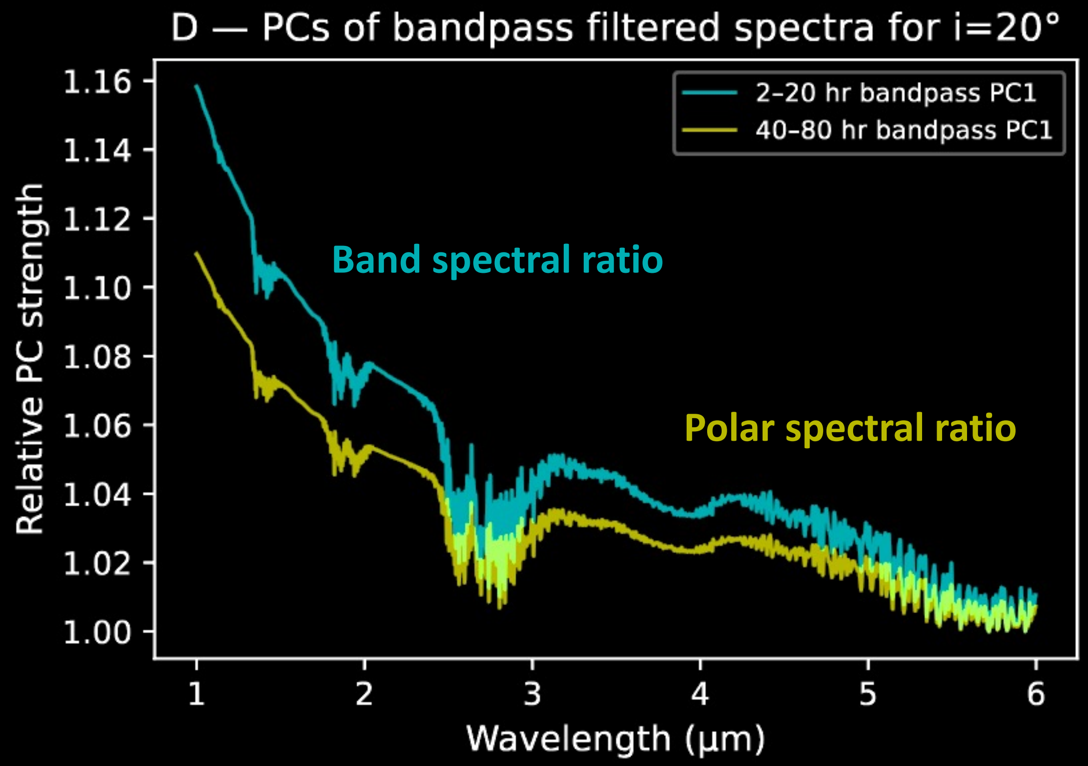
Simple methods (eg. PCA with a bandpass filter separating short vs long-period variation) can distinguish a long-period polar spectra vs. short-period band spectra ***in one BD.***

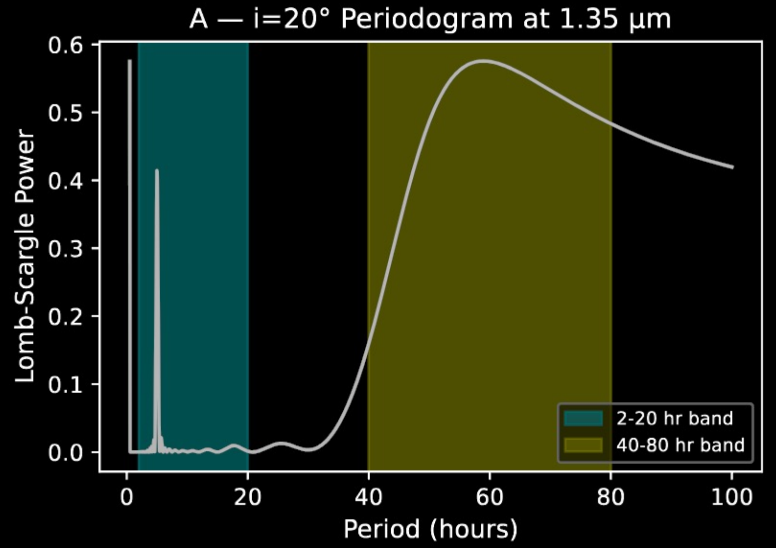
Simulated spectral timeseries w/ Maymo

(Fuda in prep. 2026)

i=30







Spectral ratio = spectra at max / min flux

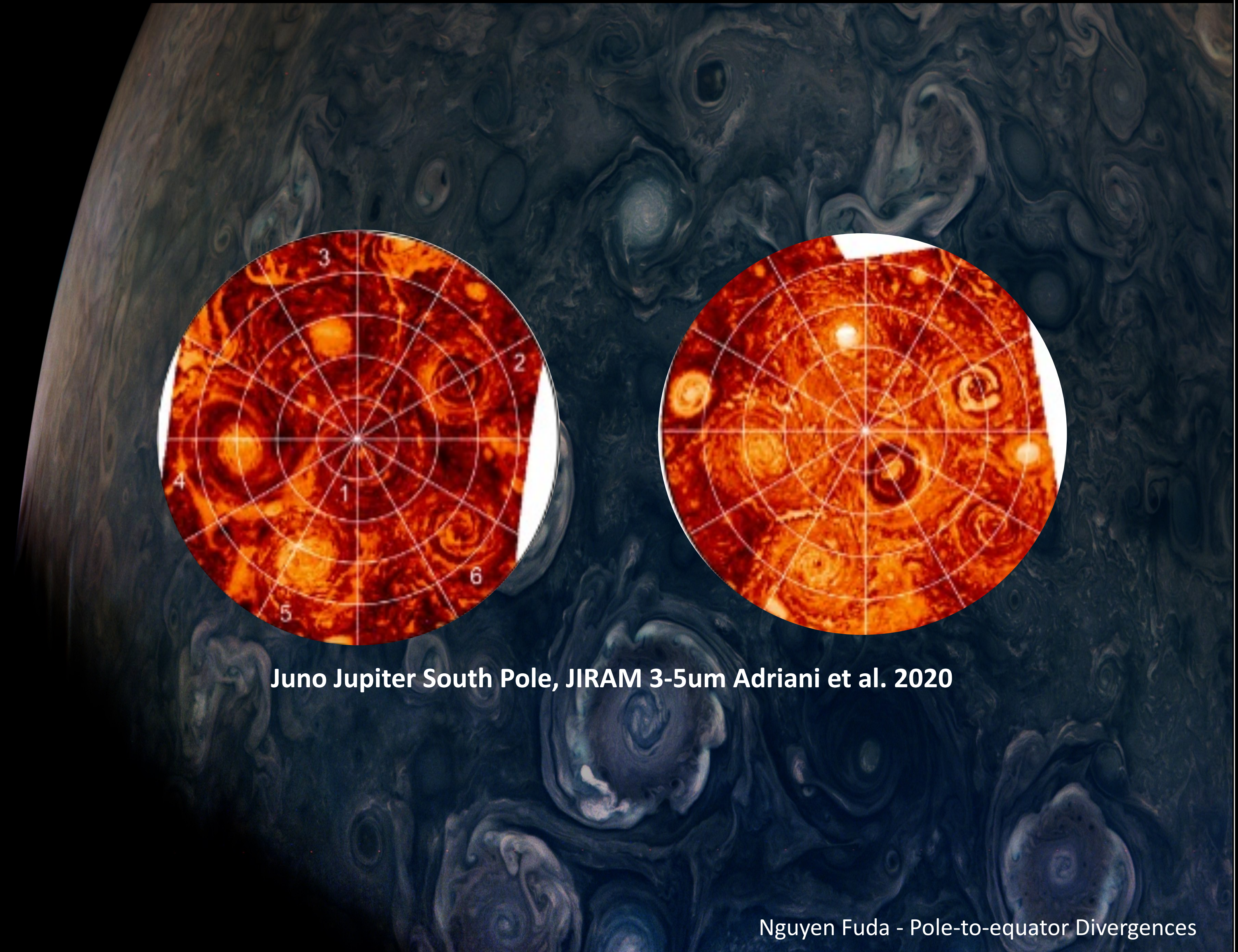
Polar ratio gets higher because more of it is visible compared to i=50.

Simple methods (eg. PCA with a bandpass filter separating short vs long-period variation) can distinguish a long-period polar spectra vs. short-period band spectra ***in one BD.***

Simulating Polar Dynamics: Inspiration from Juno/Jupiter

Aakanksha Adya,
UA Undergrad

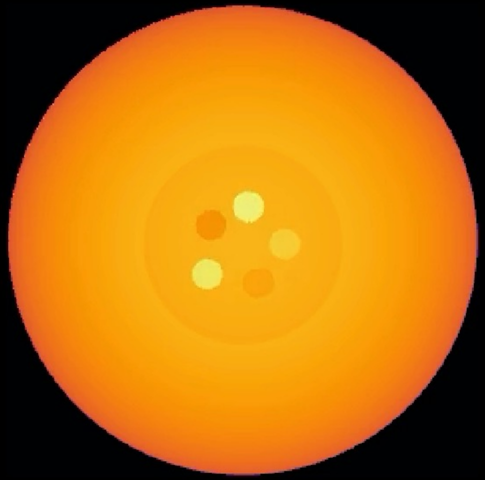
Jupiter's poles undergoes complex + dynamic evolution in cloud coverages!



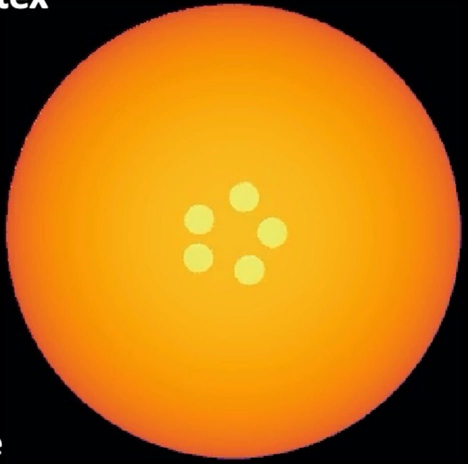
Juno Jupiter South Pole, JIRAM 3-5um Adriani et al. 2020

Nguyen Fuda - Pole-to-equator Divergences

Our model contains scenarios of polar vortex evolutions



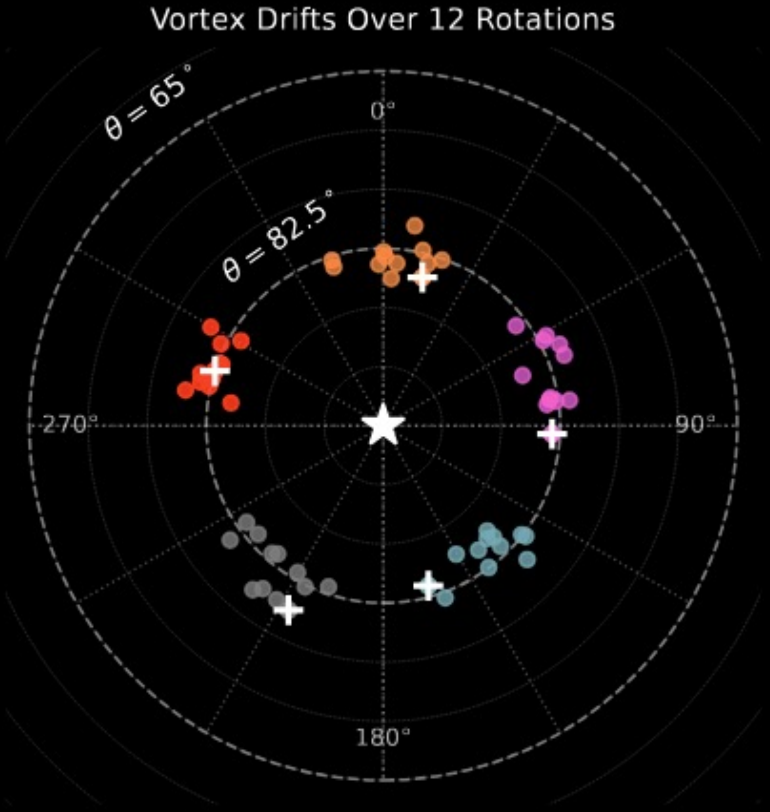
Off-phase Vortex

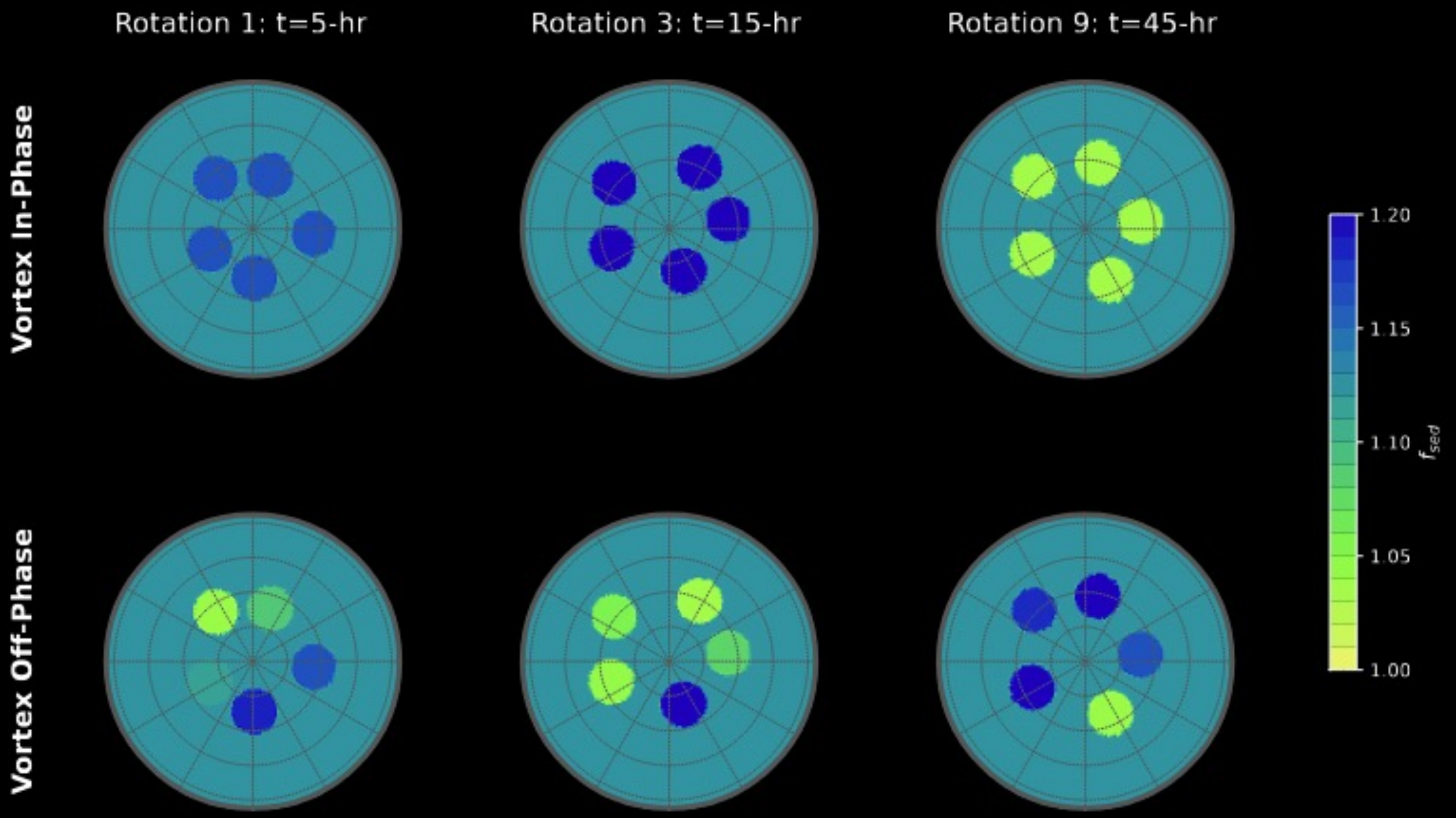


In-phase Vortex

Types of evolution create regular vs irregular light curve shapes.

Simulating Polar Dynamics: Inspiration from Juno/Jupiter





(a) Vortices centroids drift over time

(b) Polar vortices' cloud-coverages over time (polar cap unchanged)

Lightcurve Galore!

(Fuda in prep. 2026)